

Lorem Ipsum Dolor Sit Amet

Lorem Ipsum

Dolor Sit

Amet Consectetur

Lorem Ipsum University

Abstract

The task of motion planning is equivariant to rigid-body transformations, but the field the flow matching network learns is not. There are currently three techniques to restore equivariance: in the data, in the operator, or in the weights.

We ask what the task gives us before any of them. The task itself already supplies us with a frame in the workspace, by placing the origin at the midpoint of the start and goal and aligning the first axis with the direction between the two points, which removes five of the six degrees of freedom of $SE(3)$ exactly, at initialisation and at no cost. On 3D workspaces holding 40 obstacles, this raises the held-out collision-free rate from 14.6% to 51.1%. A straight line scores 15.6% on the identical problems, and the world-frame model does not beat it. A control that shows the obstacle encoder the start and goal without changing frame also reaches 15.6%, so it is the frame and not the conditioning that moves the model off the floor. One degree of freedom survives, a roll about that axis, and no continuous rule can fix it. Augmentation, frame averaging and an $SO(2)$ -equivariant architecture are worth +0.8, +0.68 and +0.15 points, and the constrained backbone reaches any level two to three times sooner, so the symmetry buys optimisation rather than accuracy. The frame is also not the largest effect available: a per-waypoint signed distance and gradient is worth +40.0 against the frame’s +36.5, and the frame still adds +24.8 alongside it.

1 Introduction

Motion planning is a fundamental problem in robotics, requiring the generation of collision-free, kinematically feasible trajectories from a start to a goal configuration in complex environments. Equivariance to rigid-body transformations is a cornerstone of generalization in this setting. It ensures that a planner solves structurally identical tasks identically, regardless of where in the world they are posed. Sampling-based [14] and optimization-based [30, 24] algorithms are equivariant by construction, since they act on the geometry directly, and they offer completeness or convergence guarantees. Their per-query computational cost limits real-time applicability. Learning-based planners [22, 8, 12, 3] amortize that cost across queries and have emerged as a promising alternative. They do not inherit the symmetry. A network trained on world coordinates relearns the same manoeuvre at every position and orientation, and

restoring equivariance in these models remains an open problem.

Current approaches place the symmetry in one of three parts of the pipeline. In the data, through augmentation. In the inference operator, through frame averaging or canonicalization [20, 13]. Or in the weights, through architectures equivariant by construction [5, 26, 9, 6], as adopted in recent robot learning systems [25, 23, 28, 29]. All three modify the model. None asks how much of the symmetry the problem statement already supplies before any model is built. We build all three mechanisms on one benchmark, hold architecture, data and training budget fixed, and measure each against what the problem supplies on its own.

A planning query names a start and a goal, and those two points determine a frame. Place the origin at their midpoint, align the first axis with the direction between them, and express every waypoint and every obstacle in that frame. Three translational and two rotational degrees of freedom vanish exactly, at initialisation, at no computational cost and with no constraint on the architecture. The six numbers that conditioned the network on the query collapse to one invariant scalar. The frame needs a second axis to be determined, and ours takes it from a fixed world axis, so rotating the whole problem rolls the frame rather than leaving it fixed. What survives is exactly that roll. $SE(3)$ collapses to $SO(2)$, not to nothing. Fig. 1 takes the group apart one stage at a time.

No continuous rule can fix the remaining roll, so it has to be handled by one of the three mechanisms above. Roll augmentation is worth +0.8 points, frame averaging +0.68, and an exactly $SO(2)$ -equivariant backbone +0.15, all seed-matched at convergence. Three unrelated ways of imposing the same symmetry recovering under a point each is a stronger statement than any one of them makes alone. The constrained architecture is not idle. It reaches any given level two to three times sooner than the unconstrained one, and it carries 19.5% more parameters, so that comparison is not confounded by capacity. The symmetry buys optimisation and not accuracy.

A collision-free rate on a new benchmark means nothing without a scale, so we measure both ends on the identical 500 held-out problems. A straight line from start to goal is collision-free on 15.6% of them. The world-frame model, trained to a plateau, reaches 14.60%. Classical planners reach 74% to 100% in under 0.4s on one CPU core. We also bound our own contribution from above. Supplying each waypoint with the signed distance to the

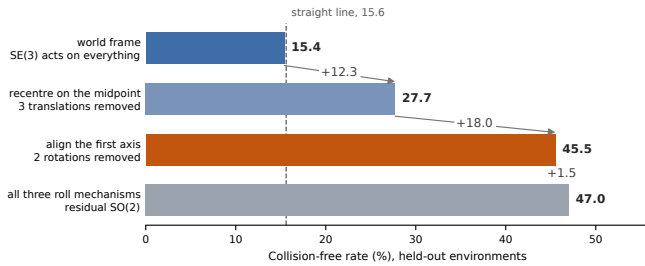


Figure 1: Lorem ipsum dolor sit amet, consectetur adipiscing elit sed do eiusmod tempor incididunt ut labore

scene and its gradient, a featurisation of data the network already receives, is worth +40.0 points where the frame is worth +36.5. The representation change studied here is not the largest effect available on this benchmark. It is not a substitute for that information either: with local geometry present the frame still adds +24.8.

Three levels appear in this paper and they are not interchangeable. The headline is 51.10 ± 0.25 against 14.60 ± 0.20 : sixty environments, trained to a plateau, three seeds, and the setting in which every mechanism comparison below is made. The 45.5 of the scaling grid runs at a fixed 20-epoch budget and answers how the gap moves with data rather than how well the method does. The 55.8 at 250 environments is the largest number we measured and rests on a single seed.

2 Related Work

The task of motion planning is equivariant to rigid-body transformations, but the field the flow matching network learns is not. There are currently three techniques to restore equivariance: in the data, in the operator, or in the weights.

We ask what the task gives us before any of them. The task itself already supplies us with a frame in the workspace, by placing the origin at the midpoint of the start and goal and aligning the first axis with the direction between the two points, which removes five of the six degrees of freedom of SE(3) exactly, at initialisation and at no cost. On 3D workspaces holding 40 obstacles, this raises the held-out collision-free rate from 14.6% to 51.1%. A straight line scores 15.6% on the identical problems, and the world-frame model does not beat it. A control that shows the obstacle encoder the start and goal without changing frame also reaches 15.6%, so it is the frame and not the conditioning that moves the model off the floor. One degree of freedom survives, a roll about that axis, and no continuous rule can fix it. Augmentation, frame averaging and an SO(2)-equivariant architecture are worth +0.8, +0.68 and +0.15 points, and the constrained backbone reaches any level two to three times sooner, so the symmetry buys optimisation rather than accuracy. The frame is also not the largest effect

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3 Method

3.1 Decomposing the group

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Proposition 1 (Lorem ipsum). *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua $X' = R_{\hat{x}}(\theta) X$.*

Proposition 2 (Dolor sit amet). *Lorem ipsum dolor sit amet consectetur adipiscing elit sed do eiusmod tempor incididunt ut labore.*

Proof. Lorem ipsum dolor sit amet consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua ut enim ad minim veniam. $\square$

4 Setup and Calibration

4.1 Benchmark and protocol

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Lorem	Ipsum (%)	Dolor	s/query
lorem ipsum	00.0	0.000	<0.001
dolor sit	00.0	0.000	0.000
amet consectetur	00.0	0.000	0.000
adipiscing elit	00.0	0.000	0.000
sed do eiusmod	00.0	0.000	0.000
tempor incididunt	00.0	−0.000	0.000
ut labore et dolore	00.0	−0.000	0.000

Table 1: Lorem ipsum dolor sit amet consectetur adipiscing elit sed do eiusmod tempor incididunt ut labore et dolore magna aliqua

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4.2 The floor, and a paired test against it

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5 Results

5.1 Scaling

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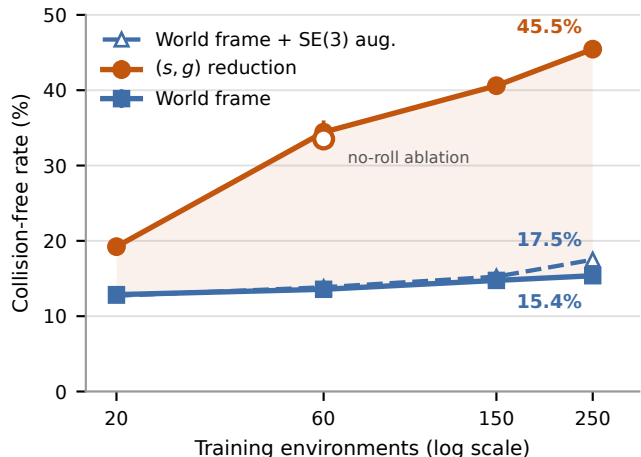


Figure 2: Lorem ipsum dolor sit amet consectetur adipiscing elit

Lorem ipsum	A	B	Dolor
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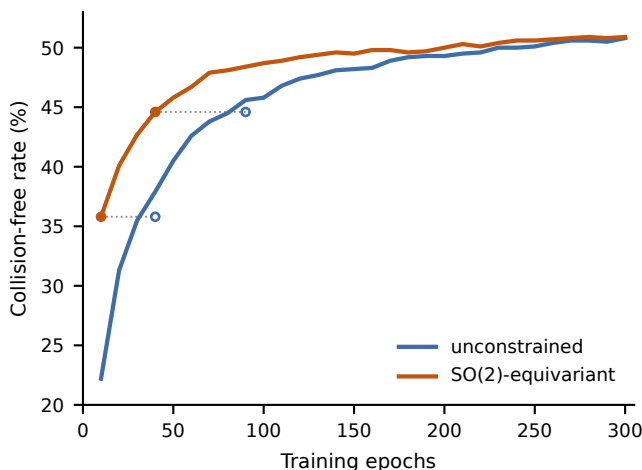


Figure 3: Lorem ipsum dolor sit amet consectetur adipiscing

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5.3 Control: local geometry, and how it scales

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5.4 Replications

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5.5 The residual, and what it bounds

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6 Limitations

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7 Conclusion

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